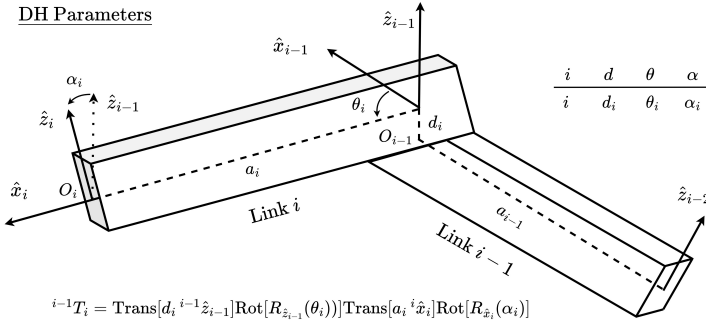


Coordinate Transforms T **Left-to-right** evaluation about the **current frame**Operators D **Right-to-left** evaluation about the **fixed frame**

$$\text{let } \begin{bmatrix} \vec{q}_1 \\ 1 \end{bmatrix} = D_1 \begin{bmatrix} \vec{p}_1 \\ 1 \end{bmatrix}, \quad \text{and we want } \begin{bmatrix} \vec{q}_2 \\ 1 \end{bmatrix} = D_0 \begin{bmatrix} \vec{p}_2 \\ 1 \end{bmatrix}$$

$$\text{then: } \begin{bmatrix} \vec{q}_2 \\ 1 \end{bmatrix} = \underbrace{{}^0T_1 D_1 {}^0T_1^{-1}}_{\text{tensor transform}} \begin{bmatrix} \vec{p}_2 \\ 1 \end{bmatrix} = D_0 \begin{bmatrix} \vec{p}_2 \\ 1 \end{bmatrix}$$

Forward KinematicsDH Parameters

$${}^{i-1}T_i = \text{Trans}[d_i \hat{z}_{i-1}] \text{Rot}[R_{\hat{z}_{i-1}}(\theta_i)] \text{Trans}[a_i \hat{x}_i] \text{Rot}[R_{\hat{x}_i}(\alpha_i)]$$

- Coordinate transform (L→R about current frame)
- If $\hat{z}_{i-1} \perp \hat{z}_i$, pick $\hat{x}_i = \pm \hat{z}_{i-1} \times \hat{z}_i$
- If $\hat{z}_{i-1} \parallel \hat{z}_i$, pick $\hat{x}_i$ to intersect O_{i-1} for kinem. calib.
- ${}^{-1}T_{\text{tool}} = {}^{-1}T_0 {}^0T_1 \dots {}^{n-1}T_n {}^nT_{\text{tool}}$
- Don't care solution keeps last row of DH zeroes except the joint variable

Inverse KinematicsFind joint parameters θ_i^* or $d_i^* \forall i$, given ${}^0\vec{d}_{0n}$ and ${}^0R_{0n}$

$${}^0\vec{d}_{0n} = \begin{bmatrix} {}^0d_{0n1} \\ {}^0d_{0n2} \\ {}^0d_{0n3} \end{bmatrix} \triangleq \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad {}^0R_{0n} \triangleq \{r_{ij}\}$$

$$\theta = 2\text{atan}\left(\pm \sqrt{\frac{1 - \cos(\theta)}{1 + \cos(\theta)}}\right) \quad \text{half angle tan formula for } \cos$$

$${}^{i-1}R_i = R_{\hat{z}_{i-1}}(\theta_i) R_{\hat{x}_i}(\alpha_i) \quad {}^{i-1}\vec{d}_{i-1,i} = d_i {}^{i-1}\hat{z}_{i-1} + a_i {}^{i-1}\hat{x}_i$$

$${}^0R_n = {}^0R_1 \dots {}^{n-1}R_n \quad {}^0T_n = {}^{-1}T_0^{-1} {}^{-1}T_{\text{tool}} {}^nT_{\text{tool}}^{-1}$$

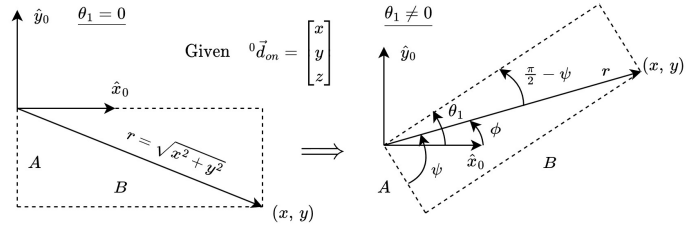
$${}^0\vec{d}_{0n} = \sum_{k=1}^n d_k {}^0R_{k-1} {}^{k-1}\hat{z}_{k-1} + a_k {}^0R_k {}^k\hat{x}_k$$

Pythagorean

- If a convenient triangle can be drawn, use a Pythagorean method to solve for joint angles
- $$c^2 = a^2 + b^2 - 2ab \cos(C) \quad \text{w/ sides } a, b, c \text{ opposite to angles } A, B, C$$

Shadow Box/Line

1. Find A, B by projecting onto a plane (usually $\hat{x}_0 - \hat{y}_0$)
2. Introduce ϕ and ψ such that $\theta_1 = \phi + (\pi/2 - \psi)$

3. Solve for $\phi = \text{atan2}(y, x)$ with **1 solution**4. Solve for $\cos(\psi) = A/r$ or $\sin(\psi) = \cos(\pi/2 + \psi) = B/r$ using the half angle tangent formula, based on what is known within A and B , resulting in **2 solutions**5. Plug into $\theta_1 = \phi + (\pi/2 - \psi)$, resulting in **2 solutions** because of ψ Projection/Influence Table

- Consider the spatial effect of perturbing each joint individually. If only a dimension (x, y, z) is affected by only one joint, project ${}^0\vec{d}_{0n}$ with that dimension, and find that joint variable

Euler Angles

$$R_{ABC} = R_A(\phi) R_B(\theta) R_C(\psi) = \{r_{ij}\} \quad A, B, C \in \{\hat{x}, \hat{y}, \hat{z}\}$$

0. Write out each rotation matrix, and simplify to total rotation matrix
1. Find first angle by taking the ratio of two r_{ij} components s.t. they simplify to $\tan(\angle_1)$, and apply half angle tan to find **2 solutions** for $\angle_1$
2. Find the r_{ij} that only includes one trig($\angle_2$). Take two other r_{ij} and multiply by $\sin/\cos$ to result in the complementary trig($\angle_2$), then use $\text{atan2}(\sin(\angle_2), \cos(\angle_2))$ to find **1 solution** for $\angle_2$
3. Similar to step 2, perform the addition cancellation twice (using four r_{ij}) to achieve estimates for $\sin(\angle_3)$ and $\cos(\angle_3)$, then $\text{atan2}(\sin(\angle_3), \cos(\angle_3))$ to find **1 solution** for $\angle_3$

If $\sin(\theta) = 0$, the rotation matrix is degenerate and becomes $R_{ABC} = R_{AC} = R_A(\phi) R_C(\psi)$. If $A = C$, it becomes $R_A(\phi + \psi)$, where ϕ and ψ cannot be uniquely solved. Convention is $\phi = 0$, and solve for θ, ψ .

Inverse Kinematic Velocities

Solve for joint (positions, velocities, accelerations) in that order, given ${}^0\dot{\vec{d}}_{0n}$ or ${}^0\vec{\omega}_{0n}$ from trajectory planner

"The time derivative of a fixed length vector is the angular velocity of the body w/ which it is fixed, crossed w/ the fixed length vector"

$$\begin{aligned} {}^{i-1}\dot{R}_i &= \dot{\theta}_i S({}^{i-1}\hat{z}_{i-1}) R_{\hat{z}}(\theta_i) \\ &= S(\dot{\theta}_i {}^{i-1}\hat{z}_{i-1}) R_{\hat{z}}(\theta_i) = {}^{i-1}\vec{\omega}_i \times R_{\hat{z}}(\theta_i) \end{aligned}$$

$$\begin{aligned}
&\text{recursive} \quad \begin{cases} \dot{\vec{d}}_{i-1,i} &= {}^0\vec{\omega}_{0i} \times {}^0\vec{d}_{i-1,i} \\ \text{rotary} \quad {}^0\vec{\omega}_{0i} &= {}^0\vec{\omega}_{0,i-1} + \dot{\theta}_i {}^0\hat{z}_{i-1} \end{cases} \\
&\text{recursive} \quad \begin{cases} \dot{\vec{d}}_{i-1,i} &= \dot{d}_i^* {}^0\hat{z}_{i-1} + {}^0\vec{\omega}_{0i} \times {}^0\vec{d}_{i-1,i} \\ \text{prismatic} \quad {}^0\vec{\omega}_{0i} &= {}^0\vec{\omega}_{0,i-1} + 0 \end{cases} \\
&\text{rotary} \quad \begin{bmatrix} {}^0\dot{\vec{d}}_{0n} \\ {}^0\dot{\vec{\omega}}_{0n} \end{bmatrix} = J_v \dot{\vec{\theta}} = \begin{bmatrix} {}^0\hat{z}_0 \cdot {}^0\vec{d}_{0n} & \dots & {}^0\hat{z}_{n-1} \cdot {}^0\vec{d}_{n-1,n} \\ {}^0\hat{z}_0 & \dots & {}^0\hat{z}_{n-1} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \vdots \\ \dot{\theta}_n \end{bmatrix} \\
&\text{prismatic} \quad \begin{bmatrix} {}^0\dot{\vec{d}}_{0n} \\ {}^0\dot{\vec{\omega}}_{0n} \end{bmatrix} = J_v \dot{\vec{d}} = \begin{bmatrix} {}^0\hat{z}_0 & \dots & {}^0\hat{z}_{n-1} \\ 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \dot{d}_1 \\ \vdots \\ \dot{d}_n \end{bmatrix} \quad \text{interweave accordingly}
\end{aligned}$$

1. Write out ${}^0\dot{\vec{d}}_{0n}$ or ${}^0\vec{\omega}_{0n}$ in terms of DH params
2. Isolate a joint rate by dotting the velocity vector with some coordinate vector.
3. Vector algebra until the joint rate $\dot{\theta}_i$ is isolated
4. Define a residual velocity vector $\vec{v}_1 = {}^0\dot{\vec{d}}_{0n} -$ term involving the solved for $\dot{\theta}_i$
5. Repeat steps 3 and 4 with new $\vec{v}_i$ definitions until all joint rates found
6. Acknowledge singularities when denominators ≈ 0

Inverse Kinematic Accelerations

$$\begin{aligned}
&\text{recursive} \quad \begin{cases} {}^0\ddot{\vec{d}}_{0i} = {}^0\ddot{\vec{d}}_{0,i-1} + {}^0\dot{\vec{\omega}}_{0i} \times {}^0\vec{d}_{i-1,i} + {}^0\vec{\omega}_{0i} \times ({}^0\vec{\omega}_{0i} \times {}^0\vec{d}_{i-1,i}) \\ {}^0\dot{\vec{\omega}}_{0i} = {}^0\dot{\vec{\omega}}_{0,i-1} + \dot{\theta}_i {}^0\hat{z}_{i-1} + \dot{\theta}_i {}^0\vec{\omega}_{0,i-1} \times {}^0\hat{z}_{i-1} \end{cases} \\
&\text{prismatic} \quad \begin{cases} {}^0\ddot{\vec{d}}_{0i} = {}^0\ddot{\vec{d}}_{0,i-1} + {}^0\dot{\vec{\omega}}_{0i} \times {}^0\vec{d}_{i-1,i} + {}^0\vec{\omega}_{0i} \times ({}^0\vec{\omega}_{0i} \times {}^0\vec{d}_{i-1,i}) \\ \quad + \ddot{d}_i {}^0\hat{z}_{i-1} + 2\dot{d}_i {}^0\vec{\omega}_{0,i-1} \times {}^0\hat{z}_{i-1} \\ {}^0\dot{\vec{\omega}}_{0i} = {}^0\dot{\vec{\omega}}_{0,i-1} + 0 \end{cases} \\
&\begin{bmatrix} {}^0\ddot{\vec{d}}_{0n} \\ {}^0\ddot{\vec{\omega}}_{0n} \end{bmatrix} = J_v \ddot{\vec{\theta}} + J_v \ddot{\vec{d}} \quad \text{where } J_v \text{ is complicated}
\end{aligned}$$

After computing the bias accel. $\vec{a} = J_v \ddot{\vec{\theta}}$, the following can be solved using the same method as inverse velocities, as "accelerating from rest looks just like velocity"

$$\underbrace{\begin{bmatrix} {}^0\ddot{\vec{d}}_{0n} \\ {}^0\ddot{\vec{\omega}}_{0n} \end{bmatrix}}_{\text{from traj}} - \underbrace{\vec{a}}_{\text{leftovers}} = \underbrace{J_v \ddot{\vec{\theta}}}_{2 \text{ dot terms}} \iff \begin{bmatrix} {}^0\ddot{\vec{d}}_{0n} \\ {}^0\ddot{\vec{\omega}}_{0n} \end{bmatrix} - \vec{a} = \sum_i \ddot{\theta}_i \text{ terms}$$

Statics

- Q_i = point of center of mass of link i
- $Q_{ik} = k^{\text{th}}$ point on link i
- $\vec{r}_{ij} = Q_j - O_i$ = center of mass of link j wrt to origin i
- $m_i = \int \int \int \rho dV$ = mass of link i
- $\vec{q}_{k,ij} = j^{\text{th}}$ point on link i wrt to origin k
- $\vec{m}_{ki} = \int \int \int \vec{q}_{k,ij} \rho dV$ = mass moment of link i about origin k

- $\vec{r}_{ij} = \vec{m}_{ki}/m_i$ = mass moment of link i about origin k scaled by mass of link i
- $\vec{f}_{ij}$ = force from body i on body j
- $\vec{n}_{ij}$ = torque from body i on body j

$$\vec{r}_{k3} = \frac{m_1 \vec{r}_{k1} + m_2 \vec{r}_{k2}}{m_1 + m_2} \quad \begin{array}{l} \vec{r}_{k3} = \text{composite center of mass} \\ \text{if body 1 + body 2 = body 3} \\ \text{about origin } k \end{array}$$

Assume COG at distal end of link to simplify calcs (torques are messier otherwise). Sum torques about proximal

$$\text{def. Gravity vec:} \quad \vec{g} = -g\Gamma \quad \Gamma \in \{\hat{x}_i, \hat{y}_i, \hat{z}_i\}$$

$$\text{def. Effort by actuator } i: \quad F_i = \hat{z}_{i-1} \cdot \vec{f}_{i-1,i} \quad \tau_i = \hat{z}_{i-1} \cdot \vec{n}_{n-1,i}$$

Recursive formulation

$$\text{link } i: \quad \sum \vec{f} = \vec{0} = \vec{f}_{i-1,i} - \vec{f}_{i,i+1} + m_1 \vec{g}$$

$$\text{link } i: \quad \sum \vec{n}_{O_{i-1}} = \vec{0} = \vec{n}_{i,i-1} - \vec{n}_{i+1,i} + \vec{r}_{i-1,i} \times m_i \vec{g} - \vec{d}_{i-1,i} \times \vec{f}_{i,i+1}$$

Dynamics

Assume MOI I_i is at distal end. Sum torques about distal end to avoid parallel axis thm.

$$\vec{p} = m_i \dot{\vec{r}}_{0i} \implies (\text{net}) \quad \vec{f}_i = m_i \ddot{\vec{r}}_{0i}$$

$$\vec{i}_{0i} = I_{0i} \vec{\omega}_{0i} \implies (\text{net, COG}) \quad \vec{n}_i = \vec{\omega}_{0i} \times I_i \vec{\omega}_{0i} + I_i \dot{\vec{\omega}}_{0i}$$

$$I_{0i} = \int \int \int dV \rho \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix} = \{I_{0,jk}\}$$

I_j = inertia of link j about frame j . I_{ik} = inertia of link i about origin/point k

$$\text{def parallel axis thm:} \quad I_{0i} = I_i + m_i (||\vec{r}_{0i}||^2 \mathbb{I} - \vec{r}_{0i} \vec{r}_{0i}^T)$$

$${}^0I_{0i} = {}^0R_i {}^iI_{0i} {}^0R_i^T \quad \text{tensor transform}$$

$$\begin{cases} \tau = \tau_g + \tau_d & \text{Effects of gravity are decoupled} \\ f = f_g + f_d & \text{from dynamics. Solve independently} \end{cases}$$

Recursive Newton-Euler

Assume joint (variables, rates, accelerations) are solved for, the joint efforts are solved using $f_i = \hat{z}_{i-1} \cdot \vec{f}_{i-1,i}$ and $\tau_i = \hat{z}_{i-1} \cdot \vec{n}_{i-1,i}$ for each link

1. Solve for the n^{th} /distal joint effort

$$\begin{aligned} \sum \vec{f} &= m_n \ddot{\vec{r}}_{0n} = \vec{f}_{n-1,n} \\ \sum \vec{n}_{O_n} &= I_n \dot{\vec{\omega}}_{0n} + \vec{\omega} \times I_n \vec{\omega}_{0n} \\ &= \vec{n}_{n-1,n} - \vec{r}_{n-1,n} \times \vec{f}_{n-1,n} \end{aligned}$$

2. Working links distal to proximal, solve recursively using

$$\begin{aligned} \sum \vec{f} &= m_i \ddot{\vec{r}}_{0i} = \vec{f}_{i-1,i} \\ \sum \vec{n}_{O_i} &= I_i \dot{\vec{\omega}}_{0i} + \vec{\omega} \times I_i \vec{\omega}_{0i} \\ &= \vec{n}_{i-1,i} - \vec{r}_{i-1,i} \times \vec{f}_{i-1,i} + \vec{r}_{ii} \times \vec{f}_{i,i+1} \end{aligned}$$

Extrassss

$${}^a R_b = \begin{bmatrix} {}^a \vec{x}_b & {}^a \vec{y}_b & {}^a \vec{z}_b \end{bmatrix} \quad {}^a \vec{p} = {}^a R_b {}^b \vec{p} \quad {}^a R_b^{-1} = {}^a R_b^T$$

$$\begin{bmatrix} {}^a \vec{p} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^a R_b & {}^a \vec{d}_{ab} \\ \vec{0}^T & 1 \end{bmatrix} \begin{bmatrix} {}^b \vec{p} \\ 1 \end{bmatrix} = {}^a T_b \begin{bmatrix} {}^b \vec{p} \\ 1 \end{bmatrix} = D_b \begin{bmatrix} {}^b \vec{p} \\ 1 \end{bmatrix}$$

$${}^a T_b = \text{Trans} \left({}^a \vec{d}_{ab} \right) \text{Rot} \left({}^a R_b \right) = \begin{bmatrix} I & {}^a \vec{d}_{ab} \\ \vec{0}^T & 1 \end{bmatrix} \begin{bmatrix} {}^a R_b & \vec{0} \\ \vec{0}^T & 1 \end{bmatrix}$$

Angle-Axis

$$R_{\hat{k}}(\theta) = \cos(\theta)I + (1 - \cos(\theta))\hat{k}\hat{k}^T + \sin(\theta)S(\hat{k}) = \{r_{ij}\}$$

$$= \begin{bmatrix} C\theta + k_1^2(1 - C\theta) & k_1 k_2(1 - C\theta) - S\theta k_3 & k_1 k_3(1 - C\theta) + S\theta k_2 \\ k_1 k_2(1 - C\theta) + S\theta k_3 & C\theta + k_2^2(1 - C\theta) & k_2 k_3(1 - C\theta) - S\theta k_1 \\ k_1 k_3(1 - C\theta) + S\theta k_3 & k_2 k_3(1 - C\theta) + S\theta k_1 & C\theta + k_3^2(1 - C\theta) \end{bmatrix}$$

$$S(\hat{k}) = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix} \quad -S(\hat{k}) = S(\hat{k})^T$$

$$\theta = \text{atan2}(S\theta, C\theta) \quad C\theta = \frac{\text{Tr}(R_{\hat{k}}(\theta)) - 1}{2}$$

$$S\theta = \pm \frac{1}{2} [(r_{32} - r_{23})^2 + (r_{13} - r_{31})^2 + (r_{21} - r_{12})^2]^{1/2}$$

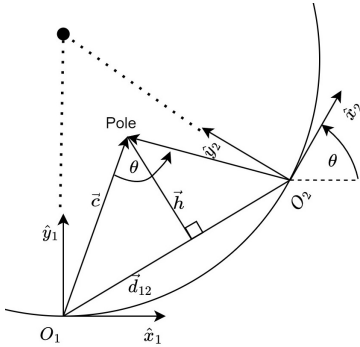
$$\hat{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \frac{1}{2S\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

Poles

$${}^a \vec{q} = R {}^a \vec{p} + \vec{d} \quad \Longleftrightarrow \quad \begin{bmatrix} {}^a \vec{q} \\ 1 \end{bmatrix} = D \begin{bmatrix} {}^a \vec{p} \\ 1 \end{bmatrix}$$

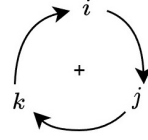
$$\vec{h} = \left(\frac{1}{2} \|\vec{d}\| \cot \left(\frac{\theta}{2} \right) \right) \left(R \left(\frac{\theta}{2} \right) \frac{\vec{d}}{\|\vec{d}\|} \right) = \|\vec{h}\| \hat{h}$$

$$\vec{c} = \frac{1}{2} \vec{d} + \vec{h} \quad \vec{c} = R\vec{c} + \vec{d} \quad // \text{Only poles satisfy}$$

Quaternions

$$\vec{q} \triangleq a + bi + cj + dk \in \mathbb{H} \quad a, b, c, d \in \mathbb{R}$$

$$\left(\vec{q} \right)^{-1} = \vec{q}^* / \|\vec{q}\| \quad i^2 = j^2 = k^2 = ijk = -1$$



$$\vec{q}\vec{v}\vec{q} = \cos \left(\frac{\theta}{2} \right) + \sin \left(\frac{\theta}{2} \right) \vec{v} \quad \vec{v} \in \mathbb{H} \setminus \mathbb{R}$$

$$\vec{p}\vec{q} = (p_0 + \vec{p})(q_0 + \vec{q}) = (\text{or imag. number math}) \\ = p_0 q_0 + p_0 \vec{q} + q_0 \vec{p} + \vec{p} \times \vec{q} - \vec{p} \cdot \vec{q}$$